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Problem and Dataset:
The goal of this project was to see whether a machine learning model could identify which LLM gave a response. The LLMs I used were ChatGPT, Gemini, and Claude. I got the data from the Idiosyncrasies in Large Language Models dataset, which contains responses generated from UltraChat prompts. I selected 500 responses from each which totalled to 1500.

Processing:
I tokenized the text using NLTK and built a vocabulary which mapped each token to an id which was a unique integer. The responses were truncated or padded to 800 tokens. I used an embedding size of 100 and a batch size of 40. I then built both a CNN and an LSTM classifier, the final CNN comprising 256 filters with kernel size 5, and the LSTM comprising hidden size 128.

Research:
Q1:
When I used LLM_Output as the input to detect what LLM generated a response I got extremely different predication accuracies from the CNN and LSTM model. For the CNN I got a much better accuracy of around between 75-85% on average for many runs, while during each of the 4 epochs, the training loss decreased heavily each line. I had to experiment with the epoch count and the number of filters. I originally had the epoch count at 5 and the total number filters at 128, with the kernel size of each filter at 3. I had an average of around 75% - 78% over many runs. I asked AI for some suggestions, and it suggested reducing the epoch count to 4 and increasing filters to 256 and kernel size to 5. Increasing the number of filters allowed the CNN model to learn even more different patterns, while increasing the kernel size to 5 allowed longer parts of the response to be examined, while 4 epochs provided a better validation accuracy showing that 5 epochs was additional training that was not much beneficial. Although my CNN model was decently accurate, my LSTM model was the opposite. The max accuracy I got was around 45%. I tried changing the epoch count from 4 to 10 and even 15, while changing the hidden state from 128 to 256. However, none of these changes improved the accuracy of the LSTM model. I asked AI to explain why the CNN model was much more accurate. The CNN performed better because the major focus on CNN is to look for recognizable patterns in the responses which the filters detected effectively, while the LSTM classified each response based on the last hidden state after processing a fixed sequence of 800 tokens. Many of the responses were padded significantly, which reduced the final representation.

Q2:
I tested the models input only, output only and input and output together. I first randomly split the individual rows, resulting in the same prompts appearing in different datasets with different LLM labels. I modified the split so that identical prompts were kept in the same training, validation or test set. After that change, the input only CNN and LSTM accuracy was around 33%. This makes sense because the same input prompts were given, there was not necessarily useful

information about which LLM generated the response, so each model just had to guess. Using the output only, CNN gave around between 75-85% accuracy while combining input and output gave around 63% the first time I ran. LSTM was much worse, as it was around 33-45% using output while remaining the same with both inputs and outputs. Overall, adding the prompt did not give the model any useful information. In fact, CNN performed worse with both input and output. For CNN, it was much better to just stick with the LLM responses as it gave much better pattern detection than just giving the prompts. Meanwhile, the LSTM was awful all around.

Q4:
To try and distinguish the characteristics of each LLM, I compared the response length, the variety of the vocabulary and the formatting. Both ChatGPT and Gemini had an average response length around 520, while Claude had much shorter, which was around 385. Claude used a more diverse set of vocabulary at around 11% unique words, while ChatGPT and Gemini were at 9%. I also compared their use of formatting symbols like -, * and #. Gemini used the most formatting symbols which was around 10%, while ChatGPT was around 9% and Claude was around 6%. These results are able to show that each LLM has their differences. Claude seemed to have the biggest difference as it gave on average shorter responses with a more diverse vocabulary and less formatting. ChatGPT and Gemini were much more similar based on these results, with Gemini edging ChatGPT in format usage.

Results:
Overall the CNN did much better than the LSTM. Using only the LLM Output CNN received between 75-85% accuracy while the LSTM was around 33-45%. With only using LLM Inputs they both got around 33% accuracy, and when both inputs and outputs were combined the CNN got around 63% and the LSTM was around 33%.

Experience:
This project introduced me to how to use both CNNs and LSTMs for classification. One of the most important things I learned is that performance can vary quite a lot with how you preprocess and tune your model. Additionally, I learned how important it is to split your data properly, as initially the same prompts were found in different datasets. Overall, the CNN performed a lot better for this data because it managed to pick up on trends in the LLMs responses better than I was able to with my implementation of the LSTM.